

Q1 - 25 June - Shift 1

If $Z = \frac{A^2 B^3}{C^4}$, then the relative error in Z will be :

(A) $\frac{\Delta A}{A} + \frac{\Delta B}{B} + \frac{\Delta C}{C}$

(B) $\frac{2\Delta A}{A} + \frac{3\Delta B}{B} - \frac{4\Delta C}{C}$

(C) $\frac{2\Delta A}{A} + \frac{3\Delta B}{B} + \frac{4\Delta C}{C}$

(D) $\frac{\Delta A}{A} + \frac{\Delta B}{B} - \frac{\Delta C}{C}$

Space for your notes:

Q2 - 25 June - Shift 1

$\vec{A}$ is a vector quantity such that $|\vec{A}| =$ non-zero constant. Which of the following expressions is true for $\vec{A}$?

(A) $\vec{A} \cdot \vec{A} = 0$

(B) $\vec{A} \times \vec{A} < 0$

(C) $\vec{A} \times \vec{A} = 0$

(D) $\vec{A} \times \vec{A} > 0$

Space for your notes:

Q3 - 25 June - Shift 1

Questions

MathonGo

Which of the following relations is true for two unit vectors $\hat{A}$ and $\hat{B}$ making an angle θ to each other?

Space for your notes:

(A) $|\hat{A} + \hat{B}| = |\hat{A} - \hat{B}| \tan \frac{\theta}{2}$

(B) $|\hat{A} - \hat{B}| = |\hat{A} + \hat{B}| \tan \frac{\theta}{2}$

(C) $|\hat{A} + \hat{B}| = |\hat{A} - \hat{B}| \cos \frac{\theta}{2}$

(D) $|\hat{A} - \hat{B}| = |\hat{A} + \hat{B}| \cos \frac{\theta}{2}$

Q4 - 25 June - Shift 2

For $z = a^2 x^3 y^{\frac{1}{2}}$, where 'a' is a constant. If percentage error in measurement of 'x' and 'y' are 4% and 12%, respectively, then the percentage error for 'z' will be %.

*Space for your notes:***Q5 - 27 June - Shift 1**

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Questions

MathonGo

A sliver wire has mass (0.6 ± 0.006) g, radius (0.5 ± 0.005) mm and length (4 ± 0.04) cm. The maximum percentage error in the measurement of its density will be :

- (A) 4% (B) 3%
(C) 6% (D) 7%

*Space for your notes:***Q6 - 27 June - Shift 2**

The distance of the Sun from earth is 1.5×10^{11} m and its angular diameter is (2000) s when observed from the earth. The diameter of the Sun will be :

- (A) 2.45×10^{10} m (B) 1.45×10^{10} m
(C) 1.45×10^9 m (D) 0.14×10^9 m

*Space for your notes:***Q7 - 29 June - Shift 1**

Two vectors $\vec{A}$ and $\vec{B}$ have equal magnitudes. If magnitude of $\vec{A} + \vec{B}$ is equal to two times the magnitude of $\vec{A} - \vec{B}$, then the angle between $\vec{A}$ and $\vec{B}$ will be :

- (A) $\sin^{-1}\left(\frac{3}{5}\right)$ (B) $\sin^{-1}\left(\frac{1}{3}\right)$
(C) $\cos^{-1}\left(\frac{3}{5}\right)$ (D) $\cos^{-1}\left(\frac{1}{3}\right)$

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Answer Key

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Q1 (C)

Q2 (C)

Q3 (B)

Q4 (18)

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Q5 (A)

Q6 (C)

Q7 (C)

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Q1 (C)

$$Z = \frac{A^2 B^3}{C^4}$$

In case of error

$$\frac{dZ}{Z} = \frac{2dA}{A} + \frac{3dB}{B} + \frac{4dC}{C}$$

$$\boxed{\frac{\Delta Z}{Z} = \frac{2\Delta A}{A} + \frac{3\Delta B}{B} + \frac{4\Delta C}{C}}$$

Q2 (C)

$$|\vec{A}| \neq 0$$

$$\vec{A} \times \vec{A} = |\vec{A}| |\vec{A}| \sin 0^\circ \hat{n} = 0$$

Q3 (B)

$$|\hat{A} + \hat{B}| = \sqrt{|\hat{A}|^2 + |\hat{B}|^2 + 2|\hat{A}||\hat{B}|\cos\theta}$$

$$= \sqrt{1+1+2\cos\theta}$$

$$= \sqrt{2(1+\cos\theta)}$$

$$= \sqrt{2 \times 2 \cos^2 \frac{\theta}{2}}$$

$$= 2 \cos \frac{\theta}{2}$$

$$|\hat{A} - \hat{B}| = \sqrt{|\hat{A}|^2 + |\hat{B}|^2 - 2|\hat{A}||\hat{B}|\cos\theta}$$

$$= \sqrt{2-2\cos\theta}$$

$$= 2 \sin \frac{\theta}{2}$$

$$\frac{|\hat{A} + \hat{B}|}{|\hat{A} - \hat{B}|} = \cot \frac{\theta}{2}$$

Q4 (18)

$$z = a^2 x^3 y^{1/2}$$

$$\frac{\Delta z}{z} = \frac{2\Delta a}{a} + \frac{3\Delta x}{x} + \frac{1}{2} \frac{\Delta y}{y}$$

a is constant

$$\frac{\Delta z}{z} \times 100 = 3(4\%) + \frac{1}{2}(12\%) = 18\%$$

Q5 (A)

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$$M = (0.6 \pm 0.006)g$$

$$r = (0.5 \pm 0.005)mm$$

$$l = (4 \pm 0.04)cm$$

$$\rho = \frac{m}{V}$$

$$\Rightarrow \frac{\Delta \rho}{\rho} = \frac{\Delta m}{m} + \frac{2\Delta r}{r} + \frac{\Delta l}{l}$$

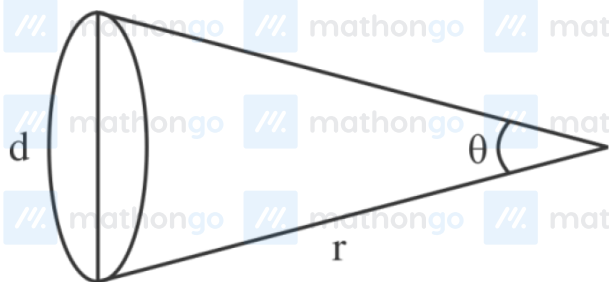
$$(Volume\ of\ cylinder = \pi r^2 l)$$

$$= \frac{0.006}{0.6} + \frac{2 \times 0.005}{0.5} + \frac{0.04}{4}$$

$$100 \times \frac{\Delta \rho}{\rho} = 4 \times 10^{-2} \times 100$$

$$\frac{\Delta \rho}{\rho} \times 100 = 4\%$$

Q6 (C)



$$\theta = \frac{d}{r}$$

$$\frac{2000}{60 \times 60} \times \frac{\pi}{180} = \frac{d}{1.5 \times 10^9}$$

$$\Rightarrow d = \frac{2000}{60 \times 60} \times \frac{\pi}{180} \times 1.5 \times 10^9$$

$$= \frac{\pi \times 1.5}{3 \times 6 \times 18} \times 10^9 = 1.45 \times 10^9$$

Q7 (C)

$$(a^2 + b^2 + 2ab \cos \theta) = 4(a^2 + b^2 - 2ab \cos \theta)$$

put $a = b$ we get

$$2a^2 + 2a^2 \cos \theta = 8a^2 - 8a^2 \cos \theta$$

$$\cos \theta = \frac{3}{5}$$